SAARTHI AUTONOMOUS INFRASTRUCTURE PROGRAMME

Phase 3 (Task 3) Technical Deliverable & Research Report: Cybersecurity (SEC-01)

Lead Engineer: Rutvika Ashtekar — **Role:** Cybersecurity Lead (Aerospace Security)

Document reference: SEC-01 Phase 3 (responds to SAARTHI-SYS-P3-TASK-002) — **Baseline Version:** 1.0 — **Date:** 24 October 2026

Executive Governance: CEO Siddharth Prasad — **Project Lead:** Adarsh Anurag — **Assistant Lead:** Deepayan Das

Repository: https://github.com/rutvikaashtekar1607/SAARTHI

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1. Executive Overview & Task Scope Confirmation

Under the SAARTHI 60-Day Engineering Lifecycle, Phase 3 (Days 19–25) is the final virtual milestone before Phase 4 hardware. The task assigned to the Cybersecurity lead is **"End-to-End V2X Telemetry Encryption Delay Models (AES-GCM / ECC over 6 Mbps / 10 Hz) & STRIDE Zero-Trust Spec."** Work package 12 asks for the cryptographic processing delay of AES-GCM and ECC over an 802.11p/C-V2X direct channel (6 Mbps PHY rate, 10 Hz / 20 Hz telemetry) "to ensure zero control loop latency degradation."

Labeling rule. Every number in this report carries one of four labels:

- **MEASURED (host):** timed on the host laptop (Intel Core i5-1135G7, Windows 11, C, Mbed TLS 3.6.7).
- **MEASURED (Renode):** instruction count read in the Renode 1.16.0 STM32F4 emulator. Renode counts instructions, not cycles.
- **MODELED:** calculated from the delay equations and the stated assumptions.
- **ESTIMATED:** Cortex-M4 times derived from Renode instruction counts with an assumed 168 MHz clock and an assumed 1.0 to 2.0 cycles per instruction (CPI).

Task Module	Core Engineering Objective	Verification Result
1. Cryptographic Latency Benchmarks	Measure AES-GCM, ECDH P-256 and ECDSA P-256 cost on the host and as Cortex-M4 instruction counts	ECDSA host: 3 runs, correctness check passed. Cortex-M4: 7 operations × 5 repeats, bench_done = 1, bench_status = 0
2. End-to-End V2X Delay Model	Model added delay for AES-GCM and ECC at 6 Mbps, 10 Hz and 20 Hz	AES-GCM fits both periods on the Cortex-M4 (ESTIMATED). Software ECDSA does not
3. Zero-Trust Specification	Extend the STRIDE analysis with zero-trust requirements	15 requirements (ZT-01 to ZT-15) with threat IDs, boundaries, verification methods and status
4. Fault-Injection Verification	Test the crypto layer under power drops, communication cuts, tampering, replay and flooding	9 of 9 tests passed; 3 of 3 negative controls showed the expected weakness (software simulation)
5. Hand-off	Repository, evidence and ICD IF-DATA	GitHub paths in Section 7; Phase 3 submission format from Smrithi pending

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2. Module 1: Cryptographic Latency Benchmarks

2.1 Host benchmarks, MEASURED (host)

Median latency per operation in µs, median of saved runs (C, Mbed TLS 3.6.7, gcc -O2). AES-GCM and ECDH come from Phase 2 (3 runs). ECDSA P-256 was added in Phase 3 (3 runs, 100 calls per repeat, 15 repeats; DER signature length 70 to 71 B; a valid signature verified and a tampered digest was rejected in every run).

Operation	Result (µs)	Range across runs (µs)
AES-128-GCM encrypt, 64 B	0.469	0.235 to 0.534
AES-128-GCM encrypt, 256 B	1.272	0.751 to 1.413

Operation	Result (µs)	Range across runs (µs)
AES-128-GCM encrypt, 1024 B	3.110	2.695 to 4.675
ECDH P-256 key generation	243.612	224.430 to 244.242
ECDH P-256 shared secret	670.224	648.928 to 709.591
ECDSA P-256 sign	263.188	262.403 to 263.235
ECDSA P-256 verify	890.081	868.598 to 906.572

2.2 Cortex-M4 instruction counts, MEASURED (Renode); times ESTIMATED

Firmware `m4_bench.c`, rebuilt from source (Phase 1 firmware was not reused): Arm GNU Toolchain 12.2.MPACBTI-Rel1 (gcc 12.2.1), Mbed TLS 3.6.7, -O2, minimal configuration `config_m4.h` (code 29,404 B text, 88 B data, 10,872 B bss). Each operation ran once as warm-up and five times between start and end markers. The run finished with bench_done = 1, bench_status = 0 and 70 markers captured.

Operation	Median instructions (min to max)	Est. time, CPI 1.0	Est. time, CPI 2.0
AES-128-GCM, 64 B	9,138 (9,138 to 9,144)	0.054 ms	0.109 ms
AES-128-GCM, 256 B	30,438 (30,438 to 30,444)	0.181 ms	0.362 ms
AES-128-GCM, 1024 B	115,638 (115,638 to 115,644)	0.688 ms	1.377 ms
ECDH P-256 key generation	4,712,615 (4,698,804 to 4,720,414)	28.051 ms	56.103 ms
ECDH P-256 shared secret	13,169,923 (13,164,342 to 13,186,541)	78.392 ms	156.785 ms
ECDSA P-256 sign	6,230,018 (6,223,693 to 6,237,141)	37.083 ms	74.167 ms
ECDSA P-256 verify	20,848,784 (20,848,784 to 20,848,787)	124.100 ms	248.200 ms

Estimate formula: **t = N × CPI / f**, with f = 168 MHz (STM32F407 clock, assumed) and CPI between 1.0 and 2.0 (assumed range). The conclusions in Section 6 hold at both CPI values.

3. Module 2: End-to-End V2X Delay Model

3.1 Governing equations

- **T_total = T_tx + T_air + T_prop + T_rx**
- **T_air = T_phy + 8 × (L + O) / R**
- **ΔT (added by security) = T_total − (T_phy + 8L/R + T_prop)**
- **T_setup = 2 × (T_keygen + T_sign + T_verify + T_shared) + 2 × (T_air(H) + T_prop)**, with H = 65 + 64 + signer-id bytes per handshake message.

L = payload (bytes), O = security overhead (bytes), R = 6 Mbps, T_phy = 40 µs, T_prop = 1 µs, and T_dec = T_enc (assumed).

Path	Overhead O	T_tx	T_rx
AES-GCM	28 B (12 B nonce + 16 B tag)	encrypt	decrypt
ECDSA	64 B signature + signer id (8 B digest or 130 B certificate, assumed)	sign	verify
AES-GCM + ECDSA	28 + 64 + id	encrypt + sign	decrypt + verify

3.2 Parameters and their source

Parameter	Value	Source
PHY rate	6 Mbps	Confirmed by Deepayan Das; stated in the Phase 3 directive
Message rates	10 Hz (100 ms) and 20 Hz (50 ms)	Phase 3 directive
PHY overhead	40 µs	Assumption (802.11p 10 MHz); does not apply to C-V2X PC5 framing
Propagation	1 µs	Assumption (about 300 m)

Parameter	Value	Source
Telemetry payload	64 B base case; 256 B and 1024 B sensitivity	Author's assumption from the state vector x_ARU; Adarsh Anurag left such assumptions to the author
Signature and signer-id sizes	64 B; 8 B or 130 B	Assumptions, to verify against IEEE 1609.2

3.3 Results, 64 B payload

Host crypto times (MEASURED), airtime MODELED:

Path	Signer id	Added delay	Total one-way	% of 100 ms	% of 50 ms
AES-GCM	n/a	38.271 µs	164.605 µs	0.165%	0.329%
ECDSA	8 B digest	1249.269 µs	1375.602 µs	1.376%	2.751%
AES-GCM + ECDSA	8 B digest	1287.540 µs	1413.874 µs	1.414%	2.828%
ECDSA	130 B cert	1411.936 µs	1538.269 µs	1.538%	3.077%
AES-GCM + ECDSA	130 B cert	1450.207 µs	1576.540 µs	1.577%	3.153%

Cortex-M4 crypto times (ESTIMATED, low = CPI 1.0, high = CPI 2.0), airtime MODELED:

Path	Signer id	Total one-way (low to high)	% of 100 ms	% of 50 ms	Fits the period?
AES-GCM	n/a	0.272 to 0.381 ms	0.27% to 0.38%	0.54% to 0.76%	Yes
ECDSA	8 B digest	161.4 to 322.6 ms	161% to 323%	323% to 645%	No
AES-GCM + ECDSA	8 B digest	161.6 to 322.8 ms	162% to 323%	323% to 646%	No
ECDSA	130 B cert	161.6 to 322.8 ms	162% to 323%	323% to 646%	No
AES-GCM + ECDSA	130 B cert	161.7 to 323.0 ms	162% to 323%	323% to 646%	No

Authenticated session setup (ECDH + ECDSA on both handshake messages): 4.58 to 4.91 ms on the host (MEASURED crypto); on the Cortex-M4, 535.7 to 536.0 ms at CPI 1.0 and 1071.0 to 1071.3 ms at CPI 2.0 (ESTIMATED).

Compute load per message period on the Cortex-M4 (ESTIMATED):

Operation	At 10 Hz (100 ms)	At 20 Hz (50 ms)
AES-GCM, 64 B	0.05% to 0.11%	0.11% to 0.22%
ECDSA sign	37% to 74%	74% to 148%
ECDSA verify	124% to 248%	248% to 496%

Results for 256 B and 1024 B payloads are in `e2e_model_output.csv`.

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4. Module 3: Zero-Trust Specification (STRIDE)

Section 14 of `01_SEC-01_STRIDE_Threat_Model.md` extends the earlier STRIDE analysis (Sections 5, 6 and 12A) for the ARU (Autonomous Roadside Unit) with five principles: no implicit trust, verify explicitly, least privilege, assume breach and fail closed, short-lived secrets.

Link-type decision. Unicast links (ARU to Command Centre) use a mutually authenticated ECDH + ECDSA session, then per-message AES-128-GCM with counter nonces; the public-key cost is paid once per session. Broadcast safety messages need per-message sender authentication, because a shared symmetric key cannot show which key holder sent a message. Because software ECDSA is too slow on the Cortex-M4 (Section 6), the broadcast path (HSM or hardware-assisted ECDSA, or a documented selective-verification policy) is an **open decision** (ZT-12).

ID	Requirement (summary)	Status
ZT-01	Authenticate every message crossing TB-06 and TB-07; drop and log failures	Verified in software simulation
ZT-02	Never trust a message by apparent sender address or radio source	Verified in software simulation

ID	Requirement (summary)	Status
ZT-03	Unicast sessions start with mutually authenticated ECDH + ECDSA, outside control-loop tasks	Open (setup time ESTIMATED 0.54 to 1.07 s)
ZT-04	Never reuse an AES-GCM nonce; counter survives power loss	Verified in software simulation; HSM counter to be confirmed (Mohan)
ZT-05	Reject replays; last accepted counter survives reboot	Verified in software simulation
ZT-06	Rotate keys before counter or NIST SP 800-38D limits and on a policy interval	Open (interval to be set)
ZT-07	Keys generated and stored in the HSM, never in plaintext on the MCU-HSM bus	Open (Mohan)
ZT-08	HSM register map locked after configuration; debug disabled	Open (Mohan)
ZT-09	Fail closed: drop, raise a security event, no unauthenticated fallback	Verified in software simulation (drop and log); safe state not tested
ZT-10	Crypto task below GNC (priority 4) and Sensing (priority 3), no shared lock	Pending (assumption; needs FreeRTOS trace)
ZT-11	Per-message crypto compute under the message period	Evidence available: AES-GCM meets it; software ECDSA does not
ZT-12	Broadcast messages use hardware or HSM-assisted ECDSA, or a documented selective-verification policy	Open decision
ZT-13	Sensor inputs untrusted until plausibility-checked; defined safe state	Open
ZT-14	Security events logged with a monotonic counter persisted across reboot	Verified in software simulation
ZT-15	Cheap pre-checks and rate limits before expensive verification	Verified in software simulation (AES-GCM path only)

5. Module 4: Fault-Injection Verification (Software Simulation)

`fault_tests.py` simulates an AES-128-GCM sender and receiver with counter nonces and injects the faults named in the Phase 3 directive (power drops, communication cuts) plus tampering, spoofing, replay and flooding. **Result: 9 of 9 tests passed, and 3 of 3 negative controls showed the expected weakness.**

Test	Fault injected	Requirement	Result
T1	None (20 valid messages)	ZT-01	PASS, 20 of 20 accepted
T2	Flipped ciphertext, tag and nonce bits	ZT-01, ZT-09	PASS, 3 of 3 rejected
T3	Wrong key, unknown sender, forged high counter	ZT-01, ZT-02, ZT-09	PASS, receiver state not advanced
T4	Replay of newest, middle and oldest message	ZT-05	PASS, 3 of 3 rejected without decryption
T5	200 sender power drops at random points	ZT-04	PASS, 22,841 nonces, 0 duplicates
T6	20 lost messages, a late old message, receiver reboot	ZT-05, ZT-09	PASS
T7	Counter rolled back	ZT-04	PASS, send blocked
T8	700 replays or unknown senders, 300 forgeries	ZT-15	PASS, 0 decryptions for the 700; forgeries capped at 50 attempts
T9	Event-log counter across reboot	ZT-14	PASS, monotonic
T5c, T6c, T7c	Volatile counters and nonce reuse (controls)	ZT-04, ZT-05	Weakness shown: 20,689 nonce reuses; replay accepted after reboot; plaintext relation leaked

Scope: this tests protocol logic in software, with a Python dictionary standing in for non-volatile memory. It does not test the HSM hardware, the radio or real flash behaviour.

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6. Pass Criteria and Conclusion on “Zero Control-Loop Latency Degradation”

“Zero” cannot be proven in general, so the following criteria were defined by the author (Adarsh Anurag left such assumptions to the author). The FreeRTOS schedule used is the one provided by Rakshita Lakkanna B.: GNC 10 ms (priority 4), Sensing 20 ms (priority 3), Communications 50 ms (priority 2), Telemetry 100 ms (priority 1).

#	Criterion	Result
C1	Crypto runs below GNC and Sensing priority and holds no shared lock	Design requirement ZT-10; **not yet verified**
C2	Crypto compute plus airtime fits inside the message period	AES-GCM: **met** (host and Cortex-M4). Software ECDSA: **not met** on the Cortex-M4
C3	Per-operation compute load stays under 100% of the period	AES-GCM: met. ECDSA sign: 37% to 148%. ECDSA verify: 124% to 496%, not met
C4	Interference from Sensing and Communications tasks included	**Not included**: their execution times are not yet measured

Conclusion. The AES-GCM path (per-message protection after an authenticated session) meets the timing criteria on the Cortex-M4, subject to C1 and C4. Software ECDSA P-256 per message does **not** fit the 10 Hz or 20 Hz periods at 168 MHz, even at the optimistic 1.0 cycles per instruction. This applies to a generic C build of Mbed TLS; optimized or hardware-accelerated ECC could be much faster, which is not measured. The recommended architecture is ECDH + AES-GCM sessions for unicast links, and HSM or hardware ECDSA (or a selective-verification policy) for broadcast safety messages. This is a design recommendation, not a measured result.

Limitations

- Cortex-M4 times are ESTIMATED. Renode is instruction-accurate, not cycle-accurate, and does not model flash wait states. The clock and CPI range are assumptions.
- Instruction counts depend on compiler and library version and apply to the build described in Section 2.2.

- The firmware uses a non-secure test random generator, for timing only. The real entropy source (assumption A-03) is not tested.
- The 40 µs PHY overhead is an 802.11p assumption. The signature and signer-id sizes need verification against IEEE 1609.2. The 64 B payload is an assumption.
- Execution times of the Sensing and Communications tasks and the WCET of a crypto task under load are not available.
- Phase 1 Renode results were not re-verified and were not used.

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7. Deliverables, GitHub Paths and Evidence

Paths are relative to the repository folder `V2X_Spec_SAARTHI`. P3 = `SEC-01 V2X Cryptographic Specification/04_Phase3_E2E_Delay_Model`.

Deliverable	Path
ECDSA host benchmark and results	`P3/bench_ecdsa.c`, `P3/results_ecdsa.csv`, `results_ecdsa_run2.csv`, `results_ecdsa_run3.csv`
End-to-end delay model and outputs	`P3/e2e_delay_model.py`, `P3/e2e_model_output.csv`, `P3/e2e_setup_output.csv`
Model assumptions	`P3/assumptions_phase3.md`
Cortex-M4 firmware	`P3/m4_bench.c`, `config_m4.h`, `linker.ld`, `startup.c`
Build, run and parse scripts	`P3/build_m4.ps1`, `P3/run_m4.resc`, `P3/parse_m4_counts.py`
Renode instruction counts	`P3/m4_estimates.csv`, `P3/m4_counts_detail.csv`
Fault-injection tests and logs	`P3/fault_tests.py`, `P3/fault_test_results.csv`, `P3/security_event_log.csv`
Zero-Trust specification	`SEC-01 V2X Cryptographic Specification/01_SEC-01_STRIDE_Threat_Model.md` (Section 14)

Deliverable	Path
Figures	`P3/e2e_added_delay.png`, `P3/m4_compute_vs_period.png`, `03_Phase2_Latency_Model/trust_boundaries.png`
Detailed report	`SEC-01 V2X Cryptographic Specification/Phase3_Report.md`
Screenshot evidence	`Evidence/Phase3_01_ECDSA_Host_Runs.png`, `Phase3_02_E2E_Model_Output.png`, `Phase3_03_M4_Build.png`, `Phase3_04_Renode_Run.png`, `Phase3_05_M4_Counts.png`, `Phase3_06_E2E_Model_M4.png`, `Phase3_07_ZeroTrust_Spec.png`, `Phase3_08_Fault_Tests.png`

Build artifacts (`bench_ecdsa.exe` , object files, `.elf` , `.map`) are excluded from the repository.

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8. Cross-Lead Collaboration, References and Open Items

Lead	Collaboration	Status
Rakshita Lakkanna B. (Real-Time Systems)	Provided FreeRTOS task periods and priorities; confirmed 10 Hz maps to the 100 ms Telemetry task; noted the 20 Hz messages map to a priority-2 task below GNC and Sensing. Sensing and Communications execution times and crypto WCET are not yet measured; she will compare against her Renode traces once estimates are shared	Periods confirmed; timing data pending

Lead	Collaboration	Status
Adarsh Anurag (GNC)	Telemetry payload size and communications budget left to the author's assumptions	Assumptions delegated
K. Mohan Krishna (Hardware Security)	Phase 2: reviewed Section 12A (TB-05, A-01 to A-04, A-09), no changes requested. Phase 3: review of HSM-related items (ZT-04, 06, 07, 08, 12)	Phase 3 review pending
Deepayan Das (Systems Integration)	Confirmed the 6 Mbps baseline rate and the 10 Hz / 20 Hz message rates (Phase 2)	Confirmed
Smrithi (ICD collation)	Format and location for Phase 3 IF-DATA submission	Pending

References.

- [Ref: Phase 3 Task Allocation Matrix & Virtual Integration Directive, SAARTHI-SYS-P3-TASK-002]
- [Ref: SEC-01 Phase 2 Report, `Phase2_Report.md`]
- [Ref: SEC-01 STRIDE Threat Model, Sections 5, 6, 12A and 14]
- [Ref: `assumptions_phase3.md`]
- [Ref: NIST SP 800-38D, Recommendation for GCM and GMAC]
- [Ref: IEEE 802.11p / IEEE 1609.2] (sizes used are assumptions to verify)

Open items. Mohan's review of the Phase 3 HSM items; Smrithi's submission format; Rakshita's Sensing, Communications and crypto-task timing; the broadcast authentication path (ZT-12); the telemetry payload size and key-rotation interval.

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9. Next-Task Recommendation

1. **HSM or hardware ECDSA offload** with Mohan's HSM: measure sign and verify latency including SPI / I2C / CAN transfer, and decide the broadcast authentication path (ZT-12).
2. **Crypto-task WCET and jitter under FreeRTOS load** in Renode with Rakshita, replacing the ESTIMATED values and closing criteria C1 and C4.

3. **Optimized ECC measurement:** repeat the Cortex-M4 count with an optimized P-256 implementation.
4. **HSM nonce-counter persistence test** across reboot and power loss (ZT-04, A-01).
5. **Agree the telemetry payload size and key-rotation interval** (ZT-06) with Adarsh and Mohan, and update the model.
6. **Phase 4 hardware:** cycle counts from the DWT counter on real STM32F4 hardware, replacing the CPI assumption.

Technical & Executive Sign-Off

Prepared by: Rutvika Ashtekar — Cybersecurity Lead (Aerospace Security)

Technical review: pending — Adarsh Anurag (Project Lead, GNC Lead), Deepayan Das (Assistant Lead)

Executive sign-off: pending — Siddharth Prasad (Chief Executive Officer)